Machine Learning Quiz (Questions 1-25) - Detailed Explanations

Q1 High variance low bias

Answer: c. High variance means overfitting. Improve generalization by using more training data or applying regularization (Ridge/Lasso).

Q2 Stratified split

Answer: d. stratify=y preserves the same class distribution in train and test sets.

Q3 Imbalanced classification

Answer: c. Precision, Recall and F1-score are more informative than accuracy.

Q4 StandardScaler

Answer: a. StandardScaler standardizes features to mean=0 and standard deviation=1.

Q5 OneHotEncoder drop='first'

Answer: b. Six categories produce five columns because one category is dropped.

Q6 Lag features and rolling means

Answer: c. They capture temporal patterns in time-series data.

Q7 High accuracy, low recall

Answer: c. Low recall means many actual positives are missed (high false negatives).

Q8 99% accuracy but misses minority

Answer: d. This indicates an imbalanced dataset.

Q9 Strong negative correlation

Answer: a. A correlation of -0.9 indicates a strong negative relationship.

Q10 Dimensionality reduction

Answer: b. t-SNE projects high-dimensional data to lower dimensions while preserving local structure.

Q11 Scale after train-test split

Answer: b. Fit the scaler only on training data to prevent data leakage.

Q12 Silhouette score

Answer: b. Measures cluster cohesion and separation.

Q13 Decision tree overfitting

Answer: d. Deep trees without pruning tend to overfit.

Q14 Log transformation

Answer: c. Reduces skewness and makes data closer to a normal distribution.

Q15 OneHotEncoder purpose

Answer: b. Converts categorical variables into binary vectors.

Q16 VIF

Answer: b. Detects multicollinearity among predictor variables.

Q17 random_state

Answer: c. Ensures reproducible train-test splits.

Q18 Classification and regression

Answer: c. Decision Trees support both tasks.

Q19 Bagging

Answer: c. Random Forest uses bootstrap sampling and builds trees in parallel.

Q20 Polynomial features

Answer: c. PolynomialFeatures() generates polynomial and interaction features.

Q21 Customer segmentation

Answer: d. Unsupervised learning is used for clustering and anomaly detection.

Q22 Train-test split

Answer: c. Evaluates how well the model generalizes to unseen data.

Q23 Regression evaluation

Answer: c. R^2 measures goodness of fit.

Q24 Imbalanced dataset

Answer: a. SMOTE and ADASYN generate synthetic minority samples.

Q25 Penalize false negatives

Answer: a. Recall should be prioritized when false negatives are costly.